APPLICATION PROGRAMMING INTERFACE SPECIFICATION

REST ENDPOINT & WEBSOCKET PROTOCOL REFERENCE

Edu Track API Reference Manual

Stateless JSON Web Token (JWT) Authentication, Role-Governed Gateways, and Real-Time Event Protocols

Base URL:	`/api/v1` or `/api` (Production Host: https://projectmernfull-stack-roan.vercel.app)
Transport Security:	HTTPS / TLS 1.3 (REST) and WSS (WebSocket)
Authentication:	HTTP Header: `Authorization: Bearer <JWT_TOKEN>`
Payload Standard:	JSON (application/json) and Multipart/form-data for uploads
Standard Response:	`{ success: boolean, data?: any, message?: string, error?: string }`

API Overview: *The Edu Track API suite provides comprehensive access to academic administration, scheduling, attendance, study material distribution, and online examination workflows. All endpoints enforce strict Role-Based Access Control (RBAC). Requests to protected routes must supply a cryptographically signed Bearer JWT token issued upon login. The server transparently executes queries across the active persistence engine (MongoDB Atlas or Fallback JSON Store).*

1. Authentication & Profile Management

POST `/api/auth/login` — Authenticate user and issue JWT token

Auth: Public | **Required Roles:** All

Request Payload / Query Params:

```
{ email: "student@edutrack.edu", password: "SecurePass123" }
```

Response Payload (HTTP 200/201):

```
{ success: true, token: "eyJhbGciOi...", user: { id: "...", name: "Pavan", role: "STUDENT", department: "CSE", section: "CSE-B" } }
```

POST `/api/auth/register` — Provision a new administrative or departmental account

Auth: Admin Only | **Required Roles:** ADMIN

Request Payload / Query Params:

```
{ fullName: "Dr. Smith", email: "smith@edutrack.edu", password: "...", role: "FACULTY", department: "CSE" }
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "User account registered successfully", userId: "usr-9021" }
```

GET `/api/auth/me` — Retrieve current authenticated user profile & session context

Auth: Bearer JWT | **Required Roles:** ADMIN, FACULTY, STUDENT

Request Payload / Query Params:

```
None (Header: Authorization: Bearer )
```

Response Payload (HTTP 200/201):

```
{ success: true, user: { id: "...", email: "...", role: "FACULTY", department: "CSE", teachingSections: ["CSE-A", "CSE-B"] } }
```

2. User Roster & Account Administration

GET `/api/users/students` — Retrieve list of students filtered by cohort

Auth: Bearer JWT | **Required Roles:** ADMIN, FACULTY

Request Payload / Query Params:

```
Query: ?department=CSE&semester;=6&section;=CSE-B
```

Response Payload (HTTP 200/201):

```
{ success: true, count: 60, students: [{ id: "...", rollNumber: "990000001", name: "Pavan", email: "p@gmail.com", attendance: 86.6 }] }
```

GET `/api/users/faculty` — Retrieve departmental teaching faculty directory

Auth: Bearer JWT | **Required Roles:** ADMIN, FACULTY

Request Payload / Query Params:

```
Query: ?department=CSE
```

Response Payload (HTTP 200/201):

```
{ success: true, faculty: [{ id: "...", name: "Prof. Maruthi Kumar", email: "m@edutrack.edu", assignedSubjects: ["CS401", "CS402"] }] }
```

POST `/api/users` — Create single student or faculty account with persistent storage

Auth: Bearer JWT | **Required Roles:** ADMIN

Request Payload / Query Params:

```
{ name: "John Doe", email: "john@edutrack.edu", role: "STUDENT", department: "CSE", semester: 6, section: "CSE-A", rollNumber: "990000088" }
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Student record stored permanently in database", student: { id: "...", ... } }
```

3. Timetable Planning & Conflict Engine

GET `/api/timetable/class` — Retrieve weekly instructional class timetable for cohort

Auth: Bearer JWT | **Required Roles:** ADMIN, FACULTY, STUDENT

Request Payload / Query Params:

```
Query: ?department=CSE&semester;=6&section;=CSE-A
```

Response Payload (HTTP 200/201):

```
{ success: true, periods: [{ dayOfWeek: "Monday", periodNumber: 1, startTime: "09:00", endTime: "10:00",
subjectCode: "CS401", facultyName: "Maruthi", room: "101" }] }
```

POST `/api/timetable/class` — Schedule class period with proactive collision validation

Auth: Bearer JWT | **Required Roles:** ADMIN

Request Payload / Query Params:

```
{ dayOfWeek: "Monday", periodNumber: 2, startTime: "10:00", endTime: "11:00", department: "CSE", semester: 6,
section: "CSE-A", subjectCode: "CS402", facultyId: "f-1", room: "Hall 101" }
```

Response Payload (HTTP 200/201):

```
Success (201): { success: true, message: "Period scheduled" } | Conflict (409): { success: false, error: "Room
Hall 101 is already booked during Monday Period 2" }
```

GET `/api/timetable/exam` — Retrieve semester examination schedule for department

Auth: Bearer JWT | **Required Roles:** ADMIN, FACULTY, STUDENT

Request Payload / Query Params:

```
Query: ?department=Computer Science & Engineering&semester;=6
```

Response Payload (HTTP 200/201):

```
{ success: true, examSchedule: [{ id: "...", examDate: "2026-11-20", startTime: "10:00 AM", endTime: "01:00 PM",
subjectCode: "CS401", courseTitle: "Distributed Systems", room: "Hall 101" }] }
```

POST `/api/timetable/exam` — Publish semester examination date and room venue

Auth: Bearer JWT | **Required Roles:** ADMIN

Request Payload / Query Params:

```
{ department: "Computer Science & Engineering", semester: 6, subjectCode: "CS401", courseTitle: "Distributed
Systems", examDate: "2026-11-20", startTime: "10:00 AM", endTime: "01:00 PM", room: "Exam Hall 101",
instructions: "University ID card mandatory." }
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Semester exam published successfully; visible to all department students" }
```

4. Daily Attendance & Course Materials

POST `/api/attendance` — Submit section attendance session (locked to today's date)

Auth: Bearer JWT | **Required Roles:** FACULTY

Request Payload / Query Params:

```
{ sessionDate: "2026-11-10", department: "CSE", semester: 6, section: "CSE-B", subjectId: "sub-401", records: [{ studentId: "s-1", status: "present" }, { studentId: "s-2", status: "absent" }] }
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Attendance logged for 60 students", timestamp: "2026-11-10T14:10:00Z" }
```

POST `/api/materials` — Upload course document scoped to teaching section

Auth: Bearer JWT (Multipart) | **Required Roles:** FACULTY

Request Payload / Query Params:

```
FormData: file: , title: "Unit 3 Notes", department: "CSE", semester: 6, section: "CSE-B", subjectId: "sub-401"
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Material uploaded successfully", material: { id: "m-91", fileUrl: "/uploads/unit3.pdf", section: "CSE-B" } }
```

GET `/api/materials` — Retrieve study materials scoped to student's section

Auth: Bearer JWT | **Required Roles:** FACULTY, STUDENT

Request Payload / Query Params:

```
Query: ?department=CSE&semester;=6&section;=CSE-B
```

Response Payload (HTTP 200/201):

```
{ success: true, materials: [{ id: "m-91", title: "Unit 3 Notes", fileName: "unit3.pdf", fileUrl: "/uploads/unit3.pdf", uploadedAt: "..." }] }
```

5. Online Examinations & Proctoring Pipeline

POST `/api/online-exams` — Create time-bounded online assessment with MCQs

Auth: Bearer JWT | **Required Roles:** FACULTY

Request Payload / Query Params:

```
{ title: "Mid-Term Assessment", department: "CSE", semester: 6, section: "CSE-B", subjectId: "sub-402", examDate: "2026-11-15", startTime: "14:00", endTime: "15:30", durationMinutes: 60, totalMarks: 20, passMarks: 12, negativeMarking: "None", questions: [{ questionText: "...", options: ["A","B","C","D"], correctOptionIndex: 1, marks: 2 }] }
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Online exam created", examId: "ex-5501" }
```

POST `/api/online-exams/:id/start` — Student begins exam; validates window & single attempt rule

Auth: Bearer JWT | **Required Roles:** STUDENT

Request Payload / Query Params:

```
Body: {} (ID in URL)
```

Response Payload (HTTP 200/201):

```
{ success: true, attemptId: "att-102", durationMinutes: 60, questions: [{ id: "q1", text: "...", options: [...] }] }
```

POST `/api/online-exams/:id/submit` — Submit exam responses, proctoring tab-switch count & auto-evaluate

Auth: Bearer JWT | **Required Roles:** STUDENT

Request Payload / Query Params:

```
{ attemptId: "att-102", answers: [{ questionId: "q1", selectedOptionIndex: 1 }], tabSwitchCount: 0 }
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Exam submitted successfully", score: 20, totalMarks: 20, isPassed: true }
```

POST `/api/online-exams/:id/release-results` — Faculty releases scores and unlocks review to students

Auth: Bearer JWT | **Required Roles:** FACULTY

Request Payload / Query Params:

```
Body: {} (ID in URL)
```

Response Payload (HTTP 200/201):

```
{ success: true, message: "Results released; exam closed for new attempts; students notified" }
```

6. WebSocket Real-Time Event Protocol (Socket.IO)

Edu Track establishes an active bi-directional WebSocket connection via Socket.IO upon user authentication.

Event Identifier	Direction	Payload Specification	Functional Action
`join_cohort`	**Client -> Server**	`{ department: "CSE", semester: 6, section: "CSE-B" }`	Subscribes socket to cohort room.
`timetable_updated`	**Server -> Client**	`{ department: "CSE", semester: 6, section: "CSE-A" }`	Triggers automatic class timetable reload.
`attendance_marked`	**Server -> Client**	`{ section: "CSE-B", subjectId: "sub-401", date: "..." }`	Updates live attendance gauge and alerts.
`material_published`	**Server -> Client**	`{ section: "CSE-B", materialTitle: "Unit 3 Notes" }`	Surfaces new study material pill in student portal.
`exam_results_released`	**Server -> Client**	`{ examId: "ex-5501", title: "Lab Exam" }`	Unlocks scorecards and review keys for test takers.
`exam_scheduled`	**Server -> Client**	`{ examId: "ex-5501", examDate: "2026-11-15" }`	Adds upcoming examination badge to student dashboard.